

Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	XXXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation
2	Proper File Structure	Files and folders are logically organized	Yes	Well organized
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Clear naming
4	Function/Method Names	Functions are descriptively named	Yes	Descriptive functions
5	Code Comments	Inline and block comments explain logic	Partial	Basic comments
6	Modular Design	Code is split into reusable functions/modules	Yes	Modular structure
7	No Redundant Code	Duplicate or unused code is removed	Yes	No duplicate code
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Error handling added

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High/Medium/Low)
1	Navigation Bar	HTML, Bootstrap	All Pages	High
2	Base Template (base.html)	Flask (Jinja2)	Home, Predict, Result, About	High
3	Stylesheet (style.css)	CSS	Entire Web Application	High
4	Prediction Function	Python (Flask)	Prediction Route	High
5	Trained XGBoost Model	Python (XGBoost)	Prediction Module	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized folder structure with clear separation of frontend, backend and ML components.
Readability	5	Code is easy to understand with meaningful names and consistent formatting.
Reusability	5	Reusable templates, CSS, JavaScript and prediction logic implemented.
Documentation / Comments	4	Key sections documented; additional comments can further improve maintainability.
Overall Score	4.8/5	Clean, modular and maintainable implementation following good coding practices.